

# ActInf Livestream #003.1 "A World Unto Itself: Human Communication as A

*Livestream | 1:26:31*

live streaming

is on okay it looks like

we are a lot so welcome everybody

to team com podcast number three we're

really

looking forward to this discussion and

we really appreciate everyone

who has come out to come speak with us

so without further ado i'm going to

share my screen and we're going to jump

into

our podcast section

so this is team com podcast

number three

welcome to team com everyone this is a

recorded and an archived live stream

that we're performing we are an

experiment

in online team communication and

learning and we are the active inference

stream for now

so people can connect with us via

twitter at inferenceactive

at this gmail address here we have a

public key based team

and also if you're watching this you

probably know but we're on youtube

at this username we really welcome

all skills backgrounds and perspectives

so please let us know how we can help

you and help help provide an on-ramp for

wherever you're coming from

and please be respectful of the rules of

online communication and dialogue

so you if you're not speaking if there's

noise in the background  
you can use the raise your hand feature  
which is on the bottom  
left of jitsi and so on  
and please provide us with feedback so  
we can be improving  
our work here's going to be the  
structure of the discussion today first  
we're going to have a  
check-in and a warm-up then we're going  
to briefly cover  
tin bergens for questions which are a  
big idea  
that help us organize some of the  
concepts that are discussed in the paper  
then we will get to discussing the paper  
we'll talk about the goals of the paper  
and the abstract we'll go through the  
road map of the paper and talk about the  
section headers and how the authors get  
from a to z  
we'll talk about each one of the figures  
and try to understand them  
and then we'll have time for questions  
throughout as well as at the end  
so in the check-in or warm-up everyone  
absolutely feel free to introduce  
yourself  
briefly and your location however you'd  
like and maybe  
for each of these questions if one or  
two people could just  
give a first thought so first warm-up  
question is  
what is communication  
people can raise their hand or just kind  
of pop corn in  
yeah go ahead jared uh yeah so

alright yeah so hi everyone my name is  
uh jared faisal i'm  
uh uh in north carolina in uh  
in the united states of america um  
yes uh for me communication is just uh  
a kind of a set of behaviors and  
inferences that  
people use to align their mental states  
it's kind of like an abstract  
i think kind of way to think about it  
but um it inherits from my advisor's  
line of thought that communication is a  
sort of tool for people aligning their  
mental states to a degree good enough  
for their purposes  
um yeah that's that's it i'd  
like to think of it as a behavior that's  
extended that uh  
is like a sort of continuous extension  
of  
uh behaviors that species across the  
animal kingdom use  
um trying to think of it in a way that's  
anything too too special  
in linguistics for instance thought of  
something very very special and unique  
and it is in a certain way but  
uh the way that i found it useful to  
think about communication is as a  
another sort of behavior in people's  
toolkits  
any other thoughts on that  
yep stefan  
yeah i would say that i agree with what  
was just said there actually and uh  
if you take like an active embodied  
approach you could say that the um  
it's an extension communication is an

extension of  
the organism into the environment but  
one which can happen  
with interactions with other organisms  
also extending into their environment  
and their niche  
cool and that kind of presages this  
second question which is  
what is special interesting or important  
about  
human communication and could be nothing  
could be  
everything um any thoughts on that one  
yeah so i have a thought that it's maybe  
not super special  
anything about human communication um my  
name is shannon  
i'm at the university of california in  
merced  
but because of the pandemic i'm at home  
in south dakota for  
the foreseeable future um but yeah i  
think  
what is special about human  
communication is that it's  
the point of view that we're coming from  
so when we're like studying  
communication across the animal kingdom  
it's always the point of comparison  
either implicitly or  
explicitly um but  
you can look at like anything from  
non-verbal communicative behaviors of  
like our primate relatives or even of  
our pets with  
us um like outside of their species or  
even  
of like chemical signals and plants um

so maybe there's just a question of  
how much of something that we want to  
call agency or intentionality  
is necessary to call communication  
important in a specific  
um like niche or  
specific animal or plant kingdom or  
something like that  
very nice and science as a special form  
of human communication it's the one that  
we have  
if nothing else and so as we kind of  
continue to build towards the paper  
how do prior beliefs so beliefs that  
communicators  
have about identity narrative or culture  
play into communication in human or  
non-human systems  
sasha um yeah just  
building on these really great previous  
answers  
i'm sasha i'm in davis california  
um but i just wanted to say  
that communication is highly  
culture and context dependent and  
in thinking about how to align  
different peoples or organisms uh kind  
of  
views of the world culture  
and context plays a big role in deciding  
on how much to align  
and whether or not that's important at  
all  
cool jared uh yeah so  
um i wanted to  
briefly say something about the second  
point as well i for me real quick the  
something that's maybe the the main

point of interest for me in terms of  
human communication specifically human  
communication is the  
the uh goals of communicators in the  
human  
case tend to be or seem to be based on  
um  
a lot of for instance developmental and  
cooperative research um  
seem to be unique in that the goal is  
the alignment of  
mental states per se um so that turns on  
gathering evidence for  
that in terms of our interaction so i  
have to get certain cues from you that  
you understand what i'm saying and so  
it's not you know  
if you grab the thing for me that i i'm  
asking you to get  
by accident say accidentally you get the  
thing that i wanted you to get  
but you're also giving me cues that you  
misunderstood what i said and so it was  
just you know fortuitous  
circumstance that you happen to get what  
i wanted even in like young kids you'll  
see them try to correct that and say  
no you got the right thing this is  
really what i wanted and so those sorts  
of  
that sort of result for instance um  
suggests that  
something that's unique is the  
alignment states per se  
um rather than just having some  
functional outcome  
which is often the case in for instance  
other non-human primates forms of

communication

um and then that turns into prior

beliefs for instance how does that

play an important role in communication

for me that's building what the

uh previous speakers said um

uh your prior beliefs

are the way or an important part of your

prior beliefs are the set of cultural

conventions that you learn

um i.e how a phrase or a

a mental state some how a a mental state

or intention should be

uh made manifest in the environment for

others to pick up on

that's a set of prior beliefs how do i

best do this something that your culture

installs in you through

all sorts of joint attentional practices

earlier in

uh earlier and earlier in in development

and that's also something that species

unique that other species

don't don't seem to engage in um is the

installation of these kinds of

culturally acquired

uh patterns and and

practices for communicating one's mental

states in a way that everyone deems

good enough for um sufficient for

current purposes i guess

cool yeah our last warm-up question

is for you what might a satisfying

formal framework for communication look

like and that might include

human and non-human it might include all

these features that people are talking

about but what would be the attributes

of a comprehensive or a formal framework  
for  
thinking about all this diversity of  
communication styles in humans and  
non-humans  
what features would that kind of a model  
be expected or desired to have  
yes stiffen i think one of the key  
things  
is it would have a spatial component in  
terms of how the dynamics happened  
in space and around space particularly  
personal space around  
the organism and one of the key things  
about humans is that we can also enfold  
narrative within itself in story we can  
and we spatially arrange things so we  
often say  
like that's right what's left what's  
ahead what's been  
so that ability to somehow  
tag or embed within the niche  
is key and then the actual language part  
is somehow  
tied to that rather than the other way  
around which is what a lot of the kind  
of  
mind-based approaches it tends to be  
cool shannon i think i'm touching up on  
that  
so having this account of how we use  
space  
in conversation it also needs to be  
flexible enough to account what sasha  
was talking about  
the way different cultures use space  
like putting the past in front of you  
which is

awkward to me but makes a lot of sense  
to other

other cultures um and also aspects of  
communication that aren't linguistic  
so non-verbal would be obvious but  
communication and

like a musical situation um where no  
not necessarily any semantic referent is  
being

called upon but all the same we know  
that we're

feeling the same beat or like we're  
really

in the same abstract

space of interacting with each other  
cool um yvonne or alex anything to add  
on any of these  
points

not by no cool i agree about

that um aspect where linguistics

are one of the modes of communication

and yet there's this spatial element

and sometimes they blur when you say do

you grasp it or can you see what i'm

talking about it's

very cross-sensory and you can type out

what you mean to say

yet we're sometimes talking about

specific things sometimes not

so let's just talk really briefly about

tin bergen's four questions which are

modeled off of aristotle's four wise

and this is a image from cap heim's

paper

from the u social insect literature and

tim bergen's four questions are kind of

like a two by two matrix

the two aspects on the left side the y

axis are whether the domain of analysis  
is contemporary happening right now or  
historical in the past  
and then proximate or ultimate are  
referring  
not to ultimate like the best but rather  
like close and far  
again kind of a spatial metaphor and the  
four  
domains though they sometimes go by  
slightly different names  
are causation which is happening  
approximately and now  
uh utility which is happening it's the  
ultimate y it's the it's the teleos  
and it's happening now as well and then  
in the proximate  
version of history so recent history we  
have development or ontogeny  
and then the ultimate why of history why  
do  
birds have wings why are we here that's  
captured by evolution and just to put a  
little bit more of a specific  
vision on this as we turn towards some  
of the figures in the paper which will  
involve birds  
this is from bateson and lalon 2013  
which was a 50-year update on tim  
bergen's original 63 paper  
and here's the same four categories of  
mechanism which is causation  
uh current utility development and  
evolution reflected by  
uh respectively the neuroscience the  
actual  
meaning of the song the developmental  
basis of the song

which is showing the songs trace change  
through time  
and then the evolutionary history of the  
trait reflected on a phylogeny  
so any points on that or can we jump  
into the paper  
cool this was just to kind of bring up  
this idea for those who are maybe coming  
from outside of a philosophy of biology  
background because it's a really awesome  
framework and it helps us get on board  
with this idea that there's going to be  
more than one answer to why  
not just because of just uh different  
opinions  
there's actually different kinds of whys  
when we're talking about why is a bird  
singing there might be multiple answers  
just fundamentally  
so the paper that we are here to talk  
about today  
and lucky enough to have jared the first  
author here for a little bit longer  
is a world unto itself human  
communication as  
active inference and what i sketched the  
two goals of the paper down as  
were to formalize cooperative  
communication as a process of active  
inference  
under the free energy principle and to  
link  
the mechanistic developmental and  
evolutionary perspectives on  
communication  
by using active inference as a process  
theory  
jared what would you say about that or

how does that reflect on what you've intended when you wrote the paper yeah  
no no i think that's  
i think that's great and um i want to put particular emphasis within the context of the cs pacific paper on the second point um  
i really was hoping to um  
yeah hone in on the idea that all these different scales of analysis are in some sense linked to one another  
um i.e like you had mentioned in the last slide you know if you want to explain any kind of behavior or inferential pattern or whatever that you see any aspect of a phenotype in in the animal kingdom that's gonna turn in every single case  
on to fully explain it on on explaining what's going on in each of these different scales of analysis and showing how they all kind of fit into a coherent poll in this case if you take the active inference framework the kind of the idea of gathering evidence for your model your evolutionarily uh favored model um that bottoms out in in  
real time uh sorts of behavioral patterns um  
that yeah i was basically trying to link these different scales  
um and show how they're all sort of part of this evidence gathering procedure  
um it emerges in this idea of cooperative communication which is something  
from my advisor mike tomasulo and i hope

that uh that this will lead to some sort  
of uh  
future more uh technical explicit uh  
formal analysis cool  
and when i see tim burgin's four  
questions i think about  
yes and because not only is that a tenet  
of  
improv and communication as we know it  
at the mechanism time scale  
to foreshadow but it's like yes there's  
evolution that's an explanation  
and there's other kinds of explanations  
that we want to be addressing as well so  
it's kind of communication  
through and through and integration  
through and through  
any other comments on the goals the  
paper before we  
get to the abstract  
okay so let's work through the abstract  
and see how the authors reflected or  
summarized  
for everyone what they did they wrote  
the purpose of this paper is to extend  
the account of human cooperative  
communication  
by proposing an integrative model of  
biobehavioral dynamics of cooperative  
communication  
our formulation is based upon active  
inference  
active inference suggests that action  
perception cycles  
operate to minimize uncertainty and  
optimize an individual's internal model  
of the world  
we propose that humans are characterized

by an evolved adaptive prior belief  
that their mental states are aligned  
with or similar to those of conspecifics  
for example that we are the same sort of  
creature inhabiting the same  
sort of niche any thoughts or  
comments on these first parts of the  
abstract  
so we're within this integrative  
framework we want to bring together tinbergen  
as well as active inference with  
hopefully an eye towards  
more formal integrations that are  
possible in the future  
the use of cooperative communication  
emerges as the principal means to gather  
evidence for this belief again that  
we're in a shared niche and that we're  
similar kinds of interactors or english  
speakers or we're interested in this  
topic  
allowing for the development of a shared  
narrative that is used to disambiguate  
interactants hidden and inferred mental  
states  
any thoughts on that part what does it  
mean to disambiguate  
the hidden and inferred mental states  
well one thought on that shannon go  
ahead oh  
yeah um so i guess it's  
not just to know what the other person  
might be thinking  
um or could say in words but also to  
know that  
you're both maybe attending to the same

reference or that you're both  
experiencing a similar emotion  
or if it's different like why is the  
other person  
sad right now um so any sort of  
experience i suppose that two people  
could be feeling it's to  
figure out what they're feeling and what  
you are feeling and  
yeah a lot of them yeah  
and we have such a rich internal  
experience that it's not  
enough just to infer the external states  
of a person oh they're sitting down  
uh big surprise we need to actually do a  
lot of work  
to disambiguate or to reduce our  
uncertainty about hidden states which  
are mental in this case  
thus by using cooperative communication  
individuals effectively attuned to a  
hermeneutic niche  
composed in part of others mental states  
and reciprocally  
attuned the niche to their own ends via  
epistemic niche construction  
uh jared what is a hermeneutic niche  
and an epistemic niche how are those  
similar ideas or different  
yeah so the idea of a hermeneutic niche  
is that you know  
when we're when we're communicating we  
both have our sort of individual  
perspective so it's you know  
um it's a an i or a me  
and a you in addition to that there's  
you know the we  
the it's our shared dialogue um

we are talking about something um at the same time of course we can decompose that into like i just said our own individual perspectives and so what i was trying to get out there um was

the idea of hermeneutics is if you all have read for instance the first and fifth 2015 duet for one paper um they talk about this hermeneutic idea where it comes from the idea of like a deep reading of a text so

if you want to really fully understand a text for instance you have to both take words

in their immediate context in a sentence in which they're composed then also relate them to the larger context of the paragraph or the the um the chapter or the whole book or whatever

um so the idea is relating these two or more different scales of analysis to one another

um and seeing it as a sort of integrative explanatory whole

um and the epistemic niche construction component uh

trying to link up with uh some of the stuff that axel constant

um has been doing on niche construction so um

we construct our niche so as to offer up very precise

uh certain affordances for us

i.e we make our niche more easily understandable

that could in this case in the instance

of  
communication across long  
stretches of intergenerational time you  
see these patterns of  
what using this terminology  
epistemic niche construction where  
um languages cultures build up i should  
say cultures build up a set of  
constructions a set of four meaning  
pairings um  
[Music]  
that if you align to that set of four  
meaning pairings and  
ontogeny makes your niche very easily  
understandable  
i.e easily inferable you can  
easily grab a hold of others hidden  
states as there were others mental  
states  
by using these sorts of uh uh very  
reliable  
um sort of epidemic tools or devices  
which are  
we talked about in the paper uh  
communicative constructions  
and and the way that you describe that  
it makes me think about the  
royal we in english and if somebody said  
well  
what am i talking about it's almost like  
wait what are you talking about if  
you're not sure what you're talking  
about what are you talking about  
but when somebody says what are we  
talking about here it's a rhetorical  
question that helps us  
have a shared informational niche and i  
agree axel's work in that respect is

excellent

this means that niche construction

enables features of the niche

to encode precise reliable cues about

the deontic

or shared value of certain action

policies e.g

the utility of using communicative

constructions to disambiguate mental

states

given expectations about shared prior

beliefs

what is deontic value in this context

jared or anyone else

maybe jared first and anyone else can

raise their hand

i think if stefan wants to go oh

have in a in your environment so um

it can be useful so it's kind of a

longer standing queue that's kind of

established in the niche

um just one thing i was thinking when i

was seeing this i think it's quite

important that they're talking about

cooperative communication here

um and that narrows it down then to

something a bit more instrumental

um it's not necessarily coordinated

communication so it's not necessarily as

action-oriented as that but there is a

sense that this is trying to get to

we're trying to achieve or trying to

cooperate towards something therefore i

want to align my mental states

because if you're not cooperating with

someone you could be just trying to work

out the hidden states of like what is it

that it's important for me to talk to

them about which may not be their mental  
state it just may be  
their state in the world at the moment  
if that makes sense which i think  
this is like you've already got some  
alignment and  
assumed i think  
right and if it's not cooperative it  
might be adversarial which kind of takes  
us to dennett's intentional stance  
where you just say whatever it is if  
it's a computer or if it's a human i'm  
just trying to win this  
little local game theory interaction and  
the paper doesn't go  
that direction it moves towards  
cooperative communication so  
jared any thoughts there yeah yeah so  
this is  
all great discussion and this um  
building on the point you just said  
about you know cooperative versus  
adversarial sorts of  
for instance communication um  
even in the case of you know i say you  
know i'm using communication  
or shared language to try to outsmart  
you or outdo you or something  
um exercising like a kind of like  
machiavellian  
intelligence i think it's been called um  
that still relies on you know the shared  
presupposition i guess you want to use  
that language of um  
a cooperative bedrock so you know the  
idea is if i'm using  
this piece of language typically it's  
for you know honest

genuine communicative goals and the idea  
for instance deception relies on  
me using you know kind of exploiting  
that shared assumption  
to my own benefit to your detriment um  
so even at least in the case of humans  
when you're you know lying or something  
um adversarial communication or whatever  
still relies on there being some shared  
set of cooperative presumptions and that  
that that is what is exploited um for  
humans when they're communicating  
um and also about the  
the deontic value that those sorts of  
ideas uh deontic or shared value that  
um the idea there is that  
we you know when in ontogeny when you  
are  
attuning to learning your culture set of  
communicative constructions  
what what these constructions are are  
in speaker's brains there's um lots of  
good work for instance from  
someone who i like joan bybee that  
suggests that these are very automatized  
automatic  
sorts of mappings between form and  
meaning that  
at least in their normal use don't  
require  
expensive heavy sorts of computations  
maybe when you're first learning them  
they do but  
the more you use them and exercise them  
in actual  
joint practice with others uh in  
aligning mental states  
they become very automatic and so the

idea of geontic value is that it  
bypasses  
uh the deontic value of policy bypasses  
all the kinds of heavy computational  
updates that  
might be required by for instance  
someone learning how to use  
constructions but  
yeah and instead provides a very  
automatic mapping between  
states so it depends how you chunk a  
construction but  
constructions become linked up in a very  
automatized way and then it can be used  
effortlessly by speakers um that's the  
idea of at least in this case the  
deontical shared value of certain action  
policies  
ie constructions cool and and using the  
niche  
info niche uh idea it's kind of like a  
tragedy of the commons  
if people pollute the environmental  
niche  
it doesn't work as well as it could and  
if people pollute the informational  
niche  
with deceptive signaling it also doesn't  
work so it's a the niche metaphor  
really works well so let's think about  
the last part of the abstract  
in turn the alignment of mental states  
which are prior beliefs  
enables the emergence of a novel  
contextualizing scale of cultural  
dynamics that encompasses the actions  
and mental states of the ensemble of  
interactants and their shared

environments

the dynamics of this contextualizing

layer of cultural organizational

feedback

across scales acts to constrain the

variability of prior expectations of the

individuals who constitute it and then

just to close it with the contribution

our theory additionally builds upon the

active inference literature by

introducing a new set of

neurobiologically plausible

computational hypotheses

for cooperative communication we

conclude with directions for future

research

any last thoughts on the abstract or can

we go to the roadmap

cool really awesome and it just shows

how much there is to unpack

in the very distilled representation of

the abstract

so and jared i know that you have to

head out at some point so feel free to

just anyone can bounce when they need to

yeah i was gonna say i'm gonna i'm gonna

head out right now actually i have to go

but thanks this was this was wonderful

thank you for uh

putting us on thank you we would love to

have you at a future conversation too

so see you later thanks cool

so the road map of the paper now now

that he's gone and we can really talk

about the paper

uh it starts with an introduction part

two

is a theoretical background and they

cover several of the key  
areas of background which are the  
evolutionary origins  
and the developmental origins of  
cooperative communication so two  
different time scales evo and devo  
often called then they take  
stock and table one summarizes some of  
the key features  
that are related to cooperative  
communication so two  
a b and c don't have to do with free  
energy they don't have to do it with  
active inference  
this is more classical understandings of  
cooperative communication section three  
is  
explicitly about active inference first  
they talk about the idea of active  
inference broadly  
and about how adaptive priors help  
organisms relate in the world and about  
the idea of alignment of  
priors within the active inference  
framework figure 1  
and figure 2 show active inference in  
the world  
as well as active inference and  
communication in the world and we're  
going to look through each one of the  
figures one by one  
they then turn to the topic of deontic  
values which are shared expectations  
about the value of policies  
so section three takes us from what is  
active inference  
and why is it important for  
communication to what is this concept

of deontic value in the context of the active inference framework then they bring it together in section four where they uh more clearly talk about human communication as active inference first they look at the dynamics of communication at the time scale of mechanism which is the individual in context as well as the dyad which is the minimum communicative unit then the dynamics of communication at the time scale of ontogeny or development and then zooming back one more level what are the dynamics of communication at the time scale of cultural evolution figure 3 is a really nice representation of the communication and coupled action perception cycle figure 4 has some examples in the bird song as well as figure five of free energy minimization in the case of improvisational duets and then figure five is a duet for one learning to synchronize through active inference they then conclude with future directions and conclusions so we're going to get from introductions to communication from an evo devo perspective through the free energy principle inspired framework of active inference to the

point where we're actually making  
useful claims about the world within  
what we've put together in sections two  
three four and then leading to future  
directions what might be interesting for  
people to study  
any thoughts on the roadmap or can we go  
to the table  
and figures cool  
so table one would anyone else like to  
give this a first summary  
i put up on the top right the remember  
the cap time  
figure just so that we remember that  
these are not uh they're not conflicting  
uh layers of analysis or scales of  
analysis they're they're all part of one  
integrated  
punnett square if you will and what  
table one does is and anyone can just  
raise their hand if they  
want to jump in but otherwise i'll just  
kind of keep the  
error alive table one summarizes some of  
these key features  
that have to do with cooperative  
communication using the terminology  
in many cases from the non-active  
inference community so that's why this  
paper is such a bridge  
from a conceptual and even a keyword  
perspective  
and these four scales of analysis which  
are reflected by  
tin bergen's four whys are the real-time  
mechanism  
and the characteristic dynamics there  
have to do with ostension

which means to show or clearly  
demarcated

joint attention which is people looking  
at the same thing

and in the human case often the the um  
whites of the eyes surrounding the  
colors part of the eye have been  
discussed maybe shannon if you're  
familiar with uh with the non-human  
primate literature

i know that the non-human primate and  
humans have quite different eyes  
and people have speculated that has to  
do with attention

um also at the real time level there's  
the

optimization of relevance which is if  
someone says okay

i i know this part of trigonometry you  
can skip through that it's not relevant  
to me

or that word wasn't relevant to me  
because i don't know what it is  
so could you could you optimize the  
relevance of your sentence

by defining the word coupled  
bidirectional information flow

as well as approximate motivation to  
align and coordinate mental states  
uh like if somebody needs something and  
you want to make sure you have the other  
person's attention

hey i'm driving and i'm thirsty can you  
get me this water bottle

if the other person's not paying  
attention then you're gonna stay thirsty  
but if the other person is paying  
attention it's the motivation that you

have to align your mental states  
so they can get you the water bottle um  
and there's other  
characteristic dynamics at the time  
scale of development  
phylogeny as well as adaptation any  
any remarks on this  
um yeah i was gonna say it makes a lot  
more sense  
that this is jared's advisor  
to understand better the ontogeny of  
communication or cooperative  
communication  
so i found this an interesting um  
and really relevant bridge to  
understanding  
the onset of cooperative communication  
in a person's life as well as comparing  
it across species  
but yeah i just thought that was a  
very interesting and irrelevant  
uh connection to think about how we  
learn to  
uh cooperate and make our uh  
internal states known and that we  
actually  
are unsatisfied until we are able to  
make our internal states known  
cool stefan  
yeah i kind of think i always find these  
these  
these the way they link with each other  
kind of interesting um  
and kind of um  
could be read different ways um because  
you could you could the real-time  
mechanism could stretch out over  
time in terms of how you think your

cooperation is going to have impact on  
your life um as an as an elder  
or as a teenager but  
in some ways i kind of wonder if  
everything in most organisms comes back  
down to those evolutionary priors  
and we're able to kind of play with  
um well even even a puppy is different  
to a full grown dog in terms of how they  
fit themselves into establishing their  
role in communication but we really do  
go outside of what our gut feeling tells  
us  
and start to sort of change the rules of  
the game so to speak um  
about causation and ontology and current  
utility  
so it's it's quite it's quite useful in  
terms of sort of generative  
discussion beyond like what this  
immediately says in a direct short-term  
interaction  
cool alex or yvonne any thoughts on that  
and and also just if you you can raise  
your hand if so um  
one other thought is the two by two kind  
of punnett square  
layout makes us think if we threw a dart  
it would be okay it's that's causation  
that's utility that's development that's  
evolution another  
spatial metaphor would be like a wave  
and a ripple on a wave  
so the big wave is evolution um and then  
there's smaller waves within that  
and so yes it totally builds on the  
evolutionary priors like you were just  
saying

and it's totally about development  
because that never stops either  
and it's totally about the mechanism  
because without mechanism you don't have  
the  
voice box or you don't have the audio  
processing so it's all of those and  
they're all happening in an  
overlapping way that's not quite  
reflected by the cleanliness of this two  
by two or even a continuum  
model cool any last thoughts on the  
table or  
can we get to the figures awesome  
so here we are in figure one and for for  
team com  
this was a uh really a great image that  
we had shared a lot before this  
discussion  
because it brings together the skeleton  
of active inference in a way which is  
this cycle that goes from internal  
states which influence  
through policy states action states  
action states influence external states  
of the world which are also influencing  
each other  
but we're not too worried about those  
external states influence our  
sensory states the retina and the  
tympanic membrane and then the sensory  
states  
allow us to update our internal model of  
the world that's also  
generative so that's the very basic  
skeleton  
of the active inference cycle here  
reflected with just

one agent in feedback with the world  
and so the the flow is always going this  
way  
but also it's happened all these things  
are happening all at once  
your legs are always engaged in an  
action state  
your retina is always involved in a  
sensory state even if your eyes are  
closed  
your internal states are always updating  
so they're all happening at once  
but we can kind of trace this causal  
diagram  
uh through time and then we also see  
represented in this image the equations  
and there is a level of  
math that is just sort of brought up in  
this paper and  
cited to other places for where people  
can learn more  
but suffice to say that each of  
these different uh sectors can be  
thought of  
as happening at various levels of  
mathematical  
description so in this center part of  
the image  
it shows like the the the minimal  
version of the  
uh math and then this is a little bit  
more  
fleshed out of an equation and then  
there are citations  
and future work to be done that really  
builds it out  
more fully so what does anyone else see  
in figure one

i suppose not quite in figure one but in something you mentioned about the parts of your body that are doing the action they're doing the sensation um is a great skeleton like you said um but then there's all of the action that's happening constantly at sometimes consciousness at the subconscious level but likes the cogs of the eyes so you're constantly having the small action to resample the environment or the like proprioceptive sensations of your legs so even within that larger cycle there's all of these much much smaller cycles happening where this perception action loop is functioning in like every part of the sensorium of your body very nice good point and this image has the sensations coming into the person and this kind of puts the black box or the markov blanket if you will around the whole person but it's actually a nested markov blanket situation so each neuron has internal states action states external states for the neuron which are other neurons other cell types and then sensations so this is something that is about the world and about people but it also exists as a skeleton whenever we want to

draw the boundary around  
any kind of uh active inference system  
and thinking about communication more  
broadly than the human case is where we  
start to see oh yeah that's right  
um honest communication is the norm  
neurons don't expect each other  
to be deceptive sure it might happen but  
that's the decimal point  
and the real framework is about how  
cooperative communication goes down  
because adversarial relationships just  
don't persist over as long  
of time durations or spatial durations  
because they break down  
almost by definition and so thinking  
about this multi-level  
in space as well as multi-time scale  
kind of tying it back to the tim bergens  
levels um or four questions and  
there's a lot to this and it's awesome  
and one of the key contributions  
of the free energy principle which have  
to do  
with all of these equations that  
potentially despite the differences in  
the specifics  
between time scales and spatial scales  
and different modes of communication  
there may be some similar equations  
underlying the dynamics  
and that would be something quite  
fantastic  
so let's build on figure one to figure  
two  
so here we go we have the same the same  
colors  
and the same layout here if you just

block

off one side of the picture and you just

look at this side for example

it would be the same thing the one

person with a sensory

uh the action states that influence the

world i guess

action and sense have been flipped here

it's the action states that influence

the world

and then comes back through sense to the

person

but the two key differences here first

off is instead of just

internal states um it's now reflecting

specifically the regime of

attention so that's salience or

attention and

it's thought it could be thought of as a

subset of the total

internal generative model of the world

by the brain but now we're talking about

what specifically the in the internal

model should attention be paid to

and then the key here and this is almost

the genesis of team com

in some senses is the world for

communicating agents is other

communicating agents

and it's a deceptively simple conclusion

or

claim because it's one thing when you

have the person in feedback with the

world

that makes it it's kind of i'm thinking

about tom hanks playing ping pong

against a wall

it's him he's he's in feedback it's a

it's a dynamical system  
but the fixedness of his opponent  
makes the uh moves that he makes very  
very  
uh specified and easy to determine  
whereas once you even get two that are  
communicating and they have different  
attention they're paying attention  
different things because they have  
different internal  
models all of a sudden you get these  
totally novel and  
improvisational dynamics that they call  
thinking through other minds  
any thoughts on this  
that was a really great tom hanks ping  
pong metaphor though for  
i mean it'd be it would be that easy if  
you were just typing to a chat box and  
then it just  
copied and pasted exactly what you wrote  
back for example it wouldn't be quite a  
conversation  
and then you could imagine speaking to a  
chatbot or something like that and  
maybe attention is playing out  
differently on the computer  
versus in your brain so the internal  
states are different but again we kind  
of already knew that the action states  
are different  
the point is the shared epistemic  
resources so the epistemic  
being related to the knowledge the cues  
for the chat box what time the message  
was sent or  
who the person who sent it was the cues  
and the resources

are a shared resource they're a shared  
niche and then  
each of those uh agents depending on  
which sense  
states make it to their regime of  
attention  
are going to update their action  
policies in a different way  
and so one other thing i really like  
about this framework for communication  
is it highlights the the necessity and  
the importance of alignment  
but not perfect alignment it actually as  
a starting place  
takes that different people are going to  
be coming from different perspectives  
it's not that there's one correct  
interpretation of what person a says  
and if person b gets it perfectly they  
succeed and if they don't get it  
perfectly they failed  
it's kind of an absolutist notion of  
communication  
and here we take a really fully  
pluralist approach to communication  
where  
you have two communicating agents they  
have different internal models of the  
world  
and the communication is happening in  
this world that they share  
in the informational and in the physical  
niche that they share  
and it's actually an action-oriented  
interaction that the agents are engaged  
in that's happening as their very  
different internal models are aligning  
and so it highlights the process of

alignment without  
being too hard on the need for a perfect  
alignment and  
that just in so many situations is a  
helpful  
starting point about where where am i at  
in this conversation  
where is my conversant in this  
conversation how are we going to align  
what is this alignment for any  
thoughts here uh  
let me try to end about maybe  
about actions because as for this model  
and everything is about alignment  
of the models and actions are like not  
side effects  
but not a primary one for example if a  
team or a company speaking about playing  
actions some kind of strategies  
or something like this  
basically it's again about alignments  
and that's why  
if a company or team want  
to develop something or make some  
research  
it's need to planning recursively  
not from actions but from  
models and alignments  
yeah and to kind of give an example that  
if we were uh  
working together on a team to build a  
car the goal isn't to  
align on what the perfect car would be  
the goal  
is the action of actually making that  
car happen  
and that's going to require us to  
synchronize on mental states but also

contribute

our unique skills and perspectives and

then to give

one more example about action um

declarative speech is an action it's an

action it's a motor behavior it has to

do with the way that muscles move so

speech

is an action this is one action that

results in speed in

sound clapping but their body actions

that result in different cues and so

placing

speech uh fundamentally as an action

state it helps us get away from this um

well you have

acts and then you have speech but of

course there are speech

acts and speech is an action and in

pure conversation where the feedback is

happening through conversation

uh it gives us that same uh

action-oriented uh model that we can

apply

to non-linguistic systems and now we're

thinking about speech

as an action in a communicating system

so we can really

build on the strengths of linguistics

like jared was talking about

without overly making it about the

specifics of any human

communication style or even a cultural

communication style because

speech is again just one of the many

multi-faceted ways

that we share information and that we

call people's attention to different

features of the world  
any thoughts on figure 2 or anything  
else any questions yeah stefan  
i think this is all this is also good  
for bringing  
um action into what language is about  
like action is out there in amongst the  
world as part of sort of interacting  
with  
rather than being these representations  
in the minds that codify everything  
so i think it's quite um it's quite  
useful to see that and see the world in  
the background  
um so yeah one thing i find in some ways  
i'll always be interested to see if  
there was a triad here to see how that  
would play out  
because because that might really bring  
that out  
um but it is interesting that the  
descriptions you were given there around  
how things are unfolding and i i didn't  
haven't quite unpacked all the equations  
but i think there's something  
useful in like there's something about  
the epistemic resources and more like  
general complex adaptive systems which  
maybe aren't entirely stable  
but kind of they're interacting and  
bounces off each other  
and the more internal stuff is much more  
noisy non-linear and  
it's a sort of achievement between like  
entities in the environment  
bouncing with each other and our kind of  
non-linear kind of free energy trying to  
understand what's that all about by

minimizing free energy so i think  
this is actually quite there's quite a  
lot that can be played around  
awesome um yeah i agree about the the  
getting more than two actors so we can  
imagine another  
yellow box up here and through action  
states it's contributing to the  
epistemic resource and we can be  
super specific let's think about this  
video chat  
so the action states are what people are  
speaking about  
and the sensory states are happening to  
me through my eyes  
as i look at the screen and through my  
ears as i'm listening and  
through the sensations that i'm  
receiving i'm updating the regime of  
attention  
when i see a little uh cue an ostentive  
cue like a little  
blue box around the speaker that's  
putting my attention  
to not just where i should be looking  
but  
conceptually what i should be paying  
attention to  
um and then it also relates to the  
hermeneutics idea that jared was  
bringing up  
and so i'm imagining this epistemic  
resource our shared informational niche  
is we're all seeing basically the same  
six boxes  
on our screen our shared informational  
resources  
are identical and we can in a computer

system even know that they're identical  
but each one of us are going to unpack  
in a different way  
what is being said and so it's like the  
book's on the table  
and someone says the meaning is in the  
book well it's not  
technically in the book it's not  
physically contained in the book that's  
a spatial metaphor that it's  
in the book and so where is the meaning  
well if the meaning  
is in the relationship between the  
regime of attention  
and the shared epistemic resources well  
then you get  
almost pluralism for free you get this  
idea that different people  
are going to by looking into the book in  
their own unique way they're going to  
unpack the book differently  
maybe one person doesn't read the  
language and so the book doesn't reduce  
their uncertainty about anything  
but for someone else who's very familiar  
with the work of the  
author potentially they can read between  
the lines  
and even see things or unpack things in  
the book that aren't  
directly there and so we get all these  
amazing  
special cases and and uh richness of  
communication  
coming out within this scaffold and so  
again just to go from figure one  
where okay this is just the minimal  
possible

active inference cycle just external  
states internal states  
and then the markov blanket is being  
pierced on the inbound  
by sense and being pierced on the  
outbound by action  
let's just zoom one layer out now we're  
looking at the dyad  
and also we can just say the dyad's not  
just their  
um dissipating energy for nothing  
they're communicating to a line on  
something  
we're here to reduce our uncertainty  
about this paper and about the free  
energy principle about  
active inference um as well as other  
goals  
and so all of this is already starting  
to come out just  
as we qualitatively look at it and i  
agree that the mathematics  
uh of complex adaptive systems and  
free energy principle provide a lot of  
those non-linear dynamics that we  
would want to see for a formal  
communicative framework  
i think it's a good time to show figure  
three this is  
the last figure in this sequence of  
three  
so here we're kind of just um specifying  
uh again just drawing out again this  
communication between two brains  
and it's one canonical loop of the  
coupled action perception cycle  
so we're keeping in mind it's always  
happening continuously

and we're just going to trace out one  
loop it's kind of like we  
inject a little dye into the river we're  
just going to look at the trace as it  
goes just through one small part to get  
a bigger  
sense about the flow so it starts with  
my idea and my idea is going to  
influence  
my behavior that my behavior is going to  
fill out or pour into that epistemic  
niche my behavior in this case  
is a vocal behavior and then through our  
shared informational niche  
my behavior is going to result i mean  
this arrow  
you can imagine putting in other steps  
intermediating here like  
going through the ear and going through  
this cortex and that cortex  
but just broadly my behavior will result  
in  
your version of my idea i said grab a  
gallon of milk  
and use you got it okay i got it a  
gallon of milk  
your version of my idea is going to  
uh result in combination this is the x  
your ideas and your version of my idea  
come together  
and that is going to influence your  
behavior then  
we're back right where we started where  
your behavior is going to say  
ah okay right they said exactly the  
thing that i wanted from the grocery  
store back to me so their behavior  
suggested that the attentional regime

was there  
that the semantic content of what i  
wanted to convey was their  
i'm pretty convinced that my  
understanding  
of what you thought i said was accurate  
aka the alignment was adequate  
doesn't need to be perfect but for the  
purposes of that communication  
the alignment was adequate and then my  
ideas  
and how i perceived your perception of  
them combined  
again in the generative model the brain  
and that's going to influence behavior i  
say okay great glad you got it here's  
five bucks  
go ahead and do this so this is just  
tracing one loop  
but already we get so many of these cool  
patterns and ways to think about how  
different sources of  
prior information like my idea your idea  
relate to the uh just in time  
information that we receive everyone has  
years and years and years of their own  
thinking  
and then someone says hey what do you  
think about this policy  
and they're combining this recent  
sensory information from the person  
asking the question  
with a really big and rich sensory  
regenerative model of the world and so  
we we can capture that short term  
information as well as these deeper  
developmental and evolutionary aspects  
of

of thought together in this type of a  
model  
and again it's just a wireframe but what  
does this  
um lead anyone to think about or what  
what might be  
uh uncertainty that some of our  
listeners would have about this  
figure one two three sequence  
yeah shannon i suppose in this figure  
one two three sequence  
um everything seems like this event  
right now  
um and what's left out is this iterative  
process  
um so like going through the  
these chains of language learners  
and as you get to the end you might get  
a simple  
construction by the time you get to the  
end  
or chains of interactions between people  
where my idea is continuously being just  
slightly adjusted by your version of my  
idea which slightly adjusts  
each other node in this figure until  
we as a dyad um sort of start to align  
on the like sort of  
final state or something of our  
interaction um  
which is brought up a lot in the paper  
but it's not  
it's not super clear just from seeing  
these these three figures sort of in a  
sequence  
yep anyone else there maybe  
sasha how do you think this relates to  
improvisation

or where improvisation takes interactors

um yeah i i quite like this figure

um adding the extra or

kind of the next logical step that um

it's about

not just how i'm expressing my idea but

what is your version

of the idea that i'm expressing um

and yeah i think it's

um yeah very relevant for

uh improvisation and i'm just thinking

of like context

uh specific clues that we get from

iterative interactions and um

i'm thinking like almost like a secret

language you know that

you'd be able to speak in code

once you did it once to the person you

would know what these words mean

kind of going forward but um

as shannon mentioned like that wouldn't

be captured in in this

um in this figure uh precisely

um or explicitly um

but to know that uh when you're in this

location

you call something by a secret code name

um these are the kinds of things that

are really important

for communication that wouldn't make

sense in another context or place in

your life

cool one other thought and i wouldn't

quite characterize active inference as

an optimistic theory i i don't know if

it's optimistic or not but

what i really like as far as a starting

point for our uh

often discordant world is it starts  
from a notion that everybody has a  
unique perspective  
and through shared resources they're  
coming into alignment  
and that's why we need intercultural  
communication and  
non-violent communication all these  
types of ideas  
whereas the other way to think about  
communication  
would be somebody has the true idea of  
what this  
ism is you know socialism is x and if  
this person has the true one or this  
book has the true one and then  
those who diverge from it it's an error  
term and so instead of thinking about an  
absolutely correct  
idea and then divergence that different  
individuals have  
whether a person is the one who holds  
the absolute truth or whether no one has  
the absolute truth and everybody's deep  
in the error term  
we can just think about meeting people  
where they are  
starting with the states that people  
actually have  
instead of um starting from some  
hypothesized perfect  
state and then trying to bring people  
into alignment there  
i think it's just a very helpful idea  
and the secret language  
and the intermodal to kind of put one  
more finish on that  
if somebody puts their um scare quotes

on the video  
uh with their fingers which is a  
non-verbal cue that suggests i'm using  
this term  
ironically or it could mean different  
things but  
importantly it might mean different  
things in different cultures  
so one might want to be very careful  
when they're using  
non-verbal or secret or culturally  
specific cues  
because if someone's not looking at your  
video square the second that you make  
those scare quotes  
or you lag out for a second in the  
context of online communication  
your semantic content will not  
update their model appropriately they  
may think that you're saying something  
that's quite the opposite of what you  
intended to say  
and so a lot of the times when people  
are talking about how to improve  
the communication throughput in online  
meetings  
and there's distraction that's kind of  
the opposite of attention there's all  
these issues that confront online  
meetings and online teams  
and people will say well use the body  
language and use nonverbal cues  
and yeah it's important to use those to  
the extent you can but also  
within this framework we can already  
start to think about some cases where  
maybe that  
wouldn't help as much as we thought

especially when we're in the challenging communication contexts like low bandwidth or different language speakers so these are the communication uh scenarios where we actually want to have useful theory we don't just want to describe the easy low-hanging fruit of two people who are twins and speak the same language and read the same books we want to talk about the professor who's speaking with a child about math and we want to get them together on the same vocabulary and cut those are the communication scenarios that really are meaningful to explore and the ones that are so natural to this skeleton but are not as easy to model in other frameworks um any other thoughts on this figure yeah stefan yeah i think it's like you just mentioned in fact that they say canonical like this feels like it makes more sense for deductive thinking that you're trying to line up because just having the brain there obviously that kind of sort of it's it's in this kind of slightly disembodied slight way and so okay i have a deducted idea and then he was like okay what is that and this this could make sense now if we're gonna ground it more in like my state

or my embodied  
um feeling about something more  
generally if i'm working in a more  
iterative or inductive way or trying to  
sort of  
feel the situation abductively i don't  
know if this  
probably would be quite how i would  
maybe one of the other diagrams is more  
appropriate but this  
may be that idea that an idea pops out  
which would probably be a deontic  
or you know deductive here's a new  
equation and now it's like tick tick  
tick tick tick you can imagine it being  
a bit more schematic like this  
nice agreed and um the five e's  
uh that are sometimes brought up in  
these papers inactive  
embodied and cultured all these ways  
that go beyond just  
idea teleporting into another idea  
that is uh it's not even like that's the  
payload of communication  
it's just we can come at it from the  
other side by thinking okay what are the  
bodies doing what is actually being  
perceived in that embodied  
setting and then yeah the ideas are  
there  
but um we could just as well have had  
two  
stick figures instead of two brains  
because or we could have had a stick  
figure over here for my behavior or  
mouth or something  
so let's go to figure four and five  
which are

closer to empirical uh figures  
and these are specifically in the  
context of birdsong  
so here's the upper part of  
figure four and what is  
happening here anyone wanna just  
describe what they  
see when they look at this  
well on the x-axis of these diagrams  
yeah sasha go ahead  
um yeah so this is looking at um  
birdsong improvisation and  
um sound is a really nice way to do that  
because  
you can have a kind of a dimensional  
reduction  
and look at just the outgoing sounds  
um instead of trying to analyze like  
body posture and interaction which is  
um quite challenging and um  
when uh  
birds sing together versus sing singing  
alone  
um we might imagine that um  
we could take the two of them singing  
alone and average it out  
or something like that that could be um  
one possible outcome  
of when we put them together um  
and then the other outcomes are that  
they improvise and create something  
completely different  
that isn't just the average of the two  
agents yes so from the caption  
it says two birds endowed with prior  
expectations  
about the hidden states generating a  
shared birdsong

narrative and i love that they called  
the sheriff bird song  
narrative um the way that it works it  
says sing for two seconds and then  
listen for a response so that's why  
on the x-axis here where we have time in  
seconds you have one  
two in blue and then listen for response  
one two listen for a response  
the posterior expectations so that's  
after the priors have been updated what  
you get are posteriors  
the posterior expectations for the first  
bird are shown in red  
and expectations for the second bird are  
shown in blue both as a function of time  
the left panel shows the chaotic and  
uncoupled dynamics when the birds cannot  
hear  
each other while the right panel shows  
the synchrony in the hidden states that  
emerges when the birds exchange sensory  
signals  
so the the left image is kind of like  
two people they're in two recording  
studios  
that are separated they can't hear each  
other and they're both singing a song  
it's a call and response song and under  
the uh model or the ecosystem where  
they're not sharing  
a shared informational niche they're in  
two isolated recording studios  
there would be no expectation that the  
turn taking would happen  
um if if someone took off their  
headphones and just was talking randomly  
there'd be no expectation of turn taking

in fact to be quite impossible and so in this

singing alone like singing alone with two of the birds who are both seeing to themselves

um you see that there's basically uh there's

there's strong discordances where the one bird is like okay i'm gonna sing now and the other one's like i'm gonna sing now

and also just um i think this is more of a schematic

but let's just think about this as like a prefrontal cortex

this is where the higher order thought is happening

and then this is like a motor cortex

this is where the speech is being generated

between um the more basal regions of the brain

and the more cognitive elements so on this left side we have the bird

is singing it's it's um infer it's

deciding when to sing and also listening for a response but it's not hearing any response so it's uncoupled

you get uncoordinated dynamics however when you

allow for the birds to sing together you see

that they can synchronize so that they can actually um

communicate and really importantly it's not enough just to have the two singers in the same recording studio or have them listening to the same

sounds they actually have to have priors  
that they want to strive towards  
alignment because if someone says  
yeah yeah yeah this is a turn-taking  
song but i don't care about that  
well then you're also going to get the  
emergence of uncoordinated behavior  
so you need the sensory coupling which  
takes place through  
the shared informational niche and the  
adaptive prior  
that the person or the bird that you're  
communicating with  
is an agent like you who also wants to  
synchronize  
because that's the cooperative part if  
you're prior about your  
your um conversions is that they don't  
want to synchronize or that they're not  
going to synchronize  
well then you might not synchronize with  
them you might not give them the opening  
to synchronize  
so when we start from this place of  
communicative communicate or  
cooperative communication that means  
we're going to need that sense  
coming to the person appropriately so  
that they can turn take  
but then at an internal generative model  
level we also want to have  
this deep prior about our communication  
that we want to synchronize  
and so a lot of the later work after  
this paper  
has gone into this more about how  
individuals  
who just want to believe or want to

reduce

uncertainty about a higher order entity

that they belong to

that is what can actually lead to group  
formation

so this is a really interesting uh  
concept

any thoughts on the top half of figure  
four or can we jump to the bottom of  
figure four

cool so here we are on the bottom of  
figure four

and on these plots though the x-axis  
is the second level expectations which  
is like what you expect to be hearing  
the first level i believe is referring  
to what um like i'm

my first level expectations of my speech  
are just what i'm generating

whereas the second level expectations is  
what do they think

um what do i think that my conversant is  
getting

and so on the x-axis of both of these  
plots are the second level expectations  
of the first bird

and the y-axis is the second level  
expectations of the second bird

so here what's being communicated is a  
number between

negative 20 and 60. but it's just a  
number line so

yeah how how many uh or what's the price  
of oil today negative 20 to 60.

we're trying to reduce our uncertainty  
about this shared

niche and what we're seeing is um  
we can just go to the right side first

the  $y$  equals  $x$  line this one to one line  
would mean the second level expectations  
of the first bird is negative 10  
and the second bird got negative 10. so  
this line  
represents points on this line represent  
where there's an alignment of  
communication  
because the expectations of the first  
bird are aligned with the expectations  
of the second bird  
and we can see it really clearly on this  
right side when they're synchronized  
that whether the value being  
communicated is negative  
10 or whether it's 50 or 60 it's a  
dynamical system so we're still getting  
a trace that kind of flows above and  
beyond the line  
but we're spending most of our time on  
alignment about what is being said  
but if we go back to this isolated case  
where there's no sensory coupling and  
there's no  
ability for even if there were a deep  
prior that said i want to be aligned  
with another bird  
we're spending as much time off of this  
 $y$  equals  $x$  line as we are spending near  
it  
because the birds are just uncoordinated  
in their singing  
and so this is just showing this is what  
happens when you don't have alignment  
and this is what happens when you have  
synchronization and a manifold  
is a lower dimensional line let's just  
go with for now

that is a dimensional compression from a higher dimension  
so the points are in very higher dimensional space but when you put them down on the two-dimensional piece of paper it makes a parabola  
or in this case it's a higher dimensional space  
of two dimensions the expectations of the two birds  
yet it exists or it at least converges towards  
this one-dimensional manifold which is just a line representing  
from negative 20 to 60 what the actual communicated state is  
so this is just another a phase space dynamical representation  
of what alignment looks like in the uncoordinated case  
and then in the coordinated case where you have sensory coupling and those deep priors for alignment any thoughts on this one  
cool let's just go to five so that we can finish the figures  
and have any last thoughts okay anyone want to take a first stab  
on figure five what is being communicated here  
so in the caption of the figure it writes  
a shows changes in the posterior expectations of an order parameter for the first bird in blue and a second bird in green  
determining the chaotic structure of the songs depicted in figure four

and so and it's as a function of the  
number of exchanges so here  
the x-axis isn't the time in seconds  
with the two seconds of singing two  
seconds of listening  
now we're gonna those that two seconds  
of singing and listening we're going to  
condense that  
into one exchange so now we're kind of  
thinking in that iterated  
um way so we're starting to get towards  
iterated communication  
here by condensing through time and  
almost looking at kairos rather than  
kronos  
and then we have this uh value of a  
of a order parameter and uh  
yes so we have an order parameter and  
what we see is that the order parameter  
starts off high  
again the green and the blue are the two  
birds  
and the green bird is starting  
starts higher organization in both of  
these traces the more dim one  
the learning alone which is these two  
dimmer traces and this the  
line represents the the mean of these  
simulations and the  
the shaded area represents the um 90  
bayesian confidence intervals just to  
clarify that  
so we can start up here this green bird  
starts at  
perfectly ordered and then the blue bird  
starts at  
0.5 so a little bit more disorder maybe  
this is like a mentoring relationship

between a more encultured bird and a younger bird who's on to genetically learning how to communicate and what we see is that when we have the ability to learn together that the order parameter stays high and we can actually increase the order of disordered agents so for example if five of us had a really good internet connection and one person was lagging a lot through cues we could help order the person's communication even if they had lack

however if we learn alone then the order parameter even though it starts very high for the green bird and a medium for the bluebird in both cases you get this exponential decay so that after 30 bouts so 30 times 4 seconds 30 exchanges 30 choruses you have zero correlation the order parameter is zero

because they started out like partially synchronized just by happenstance like you told the two singers in the two different recording studios okay start singing and initially they started just by coincidence synchronized but rapidly that decays and it decays monotonically there's no chance they're ever going to recoup alignment because they're not coordinated but again if you are coordinated you can learn together

and just like sasha was saying um when  
you have improvisation and feedback  
it's not just that the green parameter  
stays at one and the blue one converges  
toward it  
it's actually that the two converge  
together and they meet in the middle  
which is just so  
um reflective of communication and we're  
getting this just from  
birdsong this isn't at the semantic  
level this isn't  
at any other higher order level sasha  
um this also makes me think of um like  
you know the two  
birds uh teaching each other the the  
song or coming to a kind of more ordered  
state  
and mentoring relationships as well as  
teams  
um and while i think um people  
kind of intuitively understand that  
there's going to be um  
some sort of meeting in the middle  
linguistically to describe it um  
but how to rise to that highest order  
um and how to converge on something that  
is  
more ordered and not drop down to  
something that is less ordered  
um i think is an interesting question  
for teens and maybe is a little more  
straightforward in like a teaching  
relationship where um one person is  
quite  
fixed and um both people know that  
that's the fixed point to rise to um  
from the student's perspective yep and

it's kind of like when we say  
everybody can learn from everyone even  
if one person's knowledge base totally  
encompasses someone else's knowledge  
base

at the very least we're learning how to  
communicate we're learning how to learn  
so

really we can honestly say everyone can  
learn from every interaction with every  
person

and now we have a framework to really go  
into that

let's just close it out with five b and  
c

so this is reminiscent of figure four  
where again we have the expectations um

here are the second level and uh  
expectations of the first bird and then  
the expectations of that second bird  
um and again we have this  $y$  equals  $x$   
manifold that would reflect

points on the line uh being in perfect  
attunement where the first bird is  
expecting to communicate 40

and the second bird's expectation is  
also 40. so that's the uh

alignment of the internal models through  
coupled communication and

a deep prior that you want to align and  
what we see is that before learning

so now we're going to be talking about  
this learning together trace

so before learning is here so before  
learning

it's not as wacky as figure four was  
where these are like all over the place  
they're it's truly uncorrelated it's

like a cloud of points  
here we're starting with a partial  
ordering  
the green order parameters one the blue  
or the parameters point five  
so this um oval is halfway between  
perfect  
and random um and there's some math that  
we could go into  
but not today but it starts off  
partially organized and you need to kind  
of start from a place of partial  
organization for communication  
and that's actually granted by evolution  
and  
the synchronization manifold before  
learning gets tightened  
to the synchronization after learning  
where the parameters have been  
improved so that now the birds are in  
improved alignment yeah stefan  
is your hand raised yeah sorry i was  
just trying to  
yeah yeah i just thought one thing was  
interesting on the other  
is the more experienced birds um  
it's like there's still a slight dip in  
their  
coordination um as they learn as they  
as you know in their kind of um  
now maybe there's a trade-off but the  
green one still goes down a bit and is  
there maybe a sense that you need to  
relax a little bit of the  
rigidity of the pattern to enable  
some compromise or some sort  
i'm not exactly sure how they came up  
with the the way that that that model

works but  
that i just noticed that i thought  
that's kind of interesting there that  
that  
there's a slip in that and maybe that's  
just a degradation of  
degradation over time it just happens  
but there seems to be something about  
that ability to relax the rigidity of  
the model that you use totally agreed  
and i'm  
i'm thinking of a math setting where the  
equations are  
in the textbook and then one person  
might be very familiar with them one  
person has less familiarity so the green  
person is teaching math with the blue  
person and it's not teaching two that's  
that absolutist idea they're  
in a shared alignment they want to align  
around a proper understanding of math  
but the teacher person still has to be  
ready to address  
unexpected questions from a student  
which may be extremely valid and  
actually push the field forward by leaps  
and bounds  
by bringing that beginner's mind to the  
equations um and if nothing else  
the teacher might have to resort to some  
informal metaphors  
or say well this this parameter wants to  
do this that's kind of  
if you just think about it from the  
textbooks perspective it reflects a  
divergence  
of that order parameter but it's being  
used

towards this higher end which is the  
shared alignment on the concepts  
and so yeah very interesting and i i  
wonder um  
about that too um and maybe that's  
reflected by initially the blue line  
takes a  
larger excursion away from order but  
then it spends  
it then converges a little bit more so  
that's like when you have a first  
conversation with somebody  
and it's like um so where are we coming  
from you're taking more of an excursion  
towards this order to be more in the  
explore phase  
and then once you have a better sense  
about where the person's coming from you  
can enter into a more  
exploit type phase um  
i think that's um almost all for our  
time and that's all the figures of the  
paper  
so on a closing page just if anyone  
wanted to share um  
uh closing thought or if they wanted to  
share a question that they had an idea  
or a future experiment so  
anyone feel free to jump in or raise  
your hand  
uh i just want to say an idea what you  
should never sing alone  
is it a russian proverb  
yes well singing alone could be nice i  
mean solo performance can be  
beautiful and uh it makes me think about  
the  
seeing the national anthem which is

often done with a solo performance  
they might do a little bit of more vocal  
flair or they might draw out a certain  
thing  
that would totally disrupt the um if  
multiple people were trying to sing it  
but when one person is doing it they can  
have a lot more artistic license  
but if you're in a choir context where  
it has to be  
synchronized you you can't just say well  
today i'm going to sing it this way  
it's not going to sound good um  
and so there's there's there's maybe  
times to sing alone but it's not singing  
together to sing alone that's for sure  
but are they singing alone i mean if  
you're performing you're singing  
performing for so you're engaging with  
an audience of some sort  
or a recording studio which still has  
a implied audience  
very true and um they're the the um  
they have a paper this figure five  
caption it's a duet for one  
and so i think that the joke there is  
it's one  
dyad it's it's a conversation for one  
this is a conversation for one  
shared markov blanket it's a  
conversation for one informational niche  
it's one  
team com it's one conversation that's  
happening  
it's not six conversations but there are  
six perspectives happening and so that's  
where i find active inference to be such  
a powerful framework

because it helps us um really give full value and cognition to six unique perspectives coming together to one shared niche and so one fallacy would be well six shared perspectives means there's no way we're gonna synchronize that's a little bit too defeatist and then the other alternative would be well because there's one informational niche there's only one conversation and if you don't get it then you're not with a picture and that's also not right but somewhere in the middle we can have a shared niche and our own perspectives

sash um

yeah just to close it out with the the meta

uh connection um you know with the words teaching each other how to sing and coming to a higher understanding um we're trying to come to a higher understanding of active inference and it was really great to have the first author of the paper here to kind of align us and set the order so that was really nice yep yep he provided context to what we discussed and so it was very

very nice to have jared come through any other thoughts

great well that concludes um team com podcast number three

i really appreciate all of our participants as

well as all of our listeners and we  
really  
are interested in staying engaged on  
these topics and hearing people's  
feedback we want to be  
really uh inclusive and engaging with  
how we're talking about these topics  
from conceptual perspectives  
uh and more so please be in touch with  
us  
and we will have this every two weeks so  
we really  
look forward to participating with many  
of you soon  
and i'm going to terminate this live  
stream thanks again everybody  
for participating and listening  
thank you